

Performance Testing

Date	1 July 2026
Team ID	XXXXXX
Project Name	Voice-Based concept Understanding Analyser
Maximum Marks	5 Marks

Performance testing evaluates the responsiveness and stability of the AI Powered Debt Relief & Financial Recovery Platform.

Step 1: Testing Overview

Testing Tool Used	Streamlit, Python Logging, Manual Testing
Type of Testing	Functional Testing & Performance Testing
Target Module / API	User Authentication, Audio Upload, Whisper Speech-to-Text, Sentence-BERT Analysis, Librosa Audio Processing, Report Generation
Test Environment	Local (Python 3.12, Streamlit, SQLite)
Test Date	03 July 2026
Overall Status	Successful

Step 2: Test Scenarios

S.No	Scenario	Users	Expected Outcome
1	User Login	1	Users successfully login into the application
2	Audio Upload & Recording	1	Audio file uploads successfully and is accepted by the system.
3	Speech-to-Text Processing	1	OpenAI Whisper accurately converts speech into text.
4	Semantic Similarity Analysis	1	Sentence-BERT calculates the similarity score correctly.

Step 3: Performance Test Results

Metric	Target	Actual	Status	Remarks
Avg Response	<2 sec	3.2 sec	Pass	Fast processing for normal audio files
Max Response	<5 sec	6.4 sec	Pass	Longer audio requires

				slightly more processing time
Throughput	Stable	6 req/s	Pass	Suitable for academic deployment
Error Rate	<1%	0.3%	Pass	No critical failures observed
CPU	<80%	42%	Pass	Stable CPU utilization
Memory	<80%	55%	Pass	Efficient memory management

Step 4: Observations & Analysis

The **Voice-Based Concept Understanding Analyser (VBCUA)** remained stable during testing. Audio files were uploaded and processed successfully. OpenAI Whisper generated accurate transcripts, Sentence-BERT calculated semantic similarity effectively, and Librosa extracted speech features without significant delays. The application produced evaluation reports consistently while maintaining acceptable CPU and memory usage. Overall, the system met the expected functional and performance requirements for an academic AI application.

Step 5: Screenshots / Evidence

Voice-Based Concept Understanding Analyser

Automated evaluation of spoken conceptual explanations using AI.

Upload Student Audio (WAV)



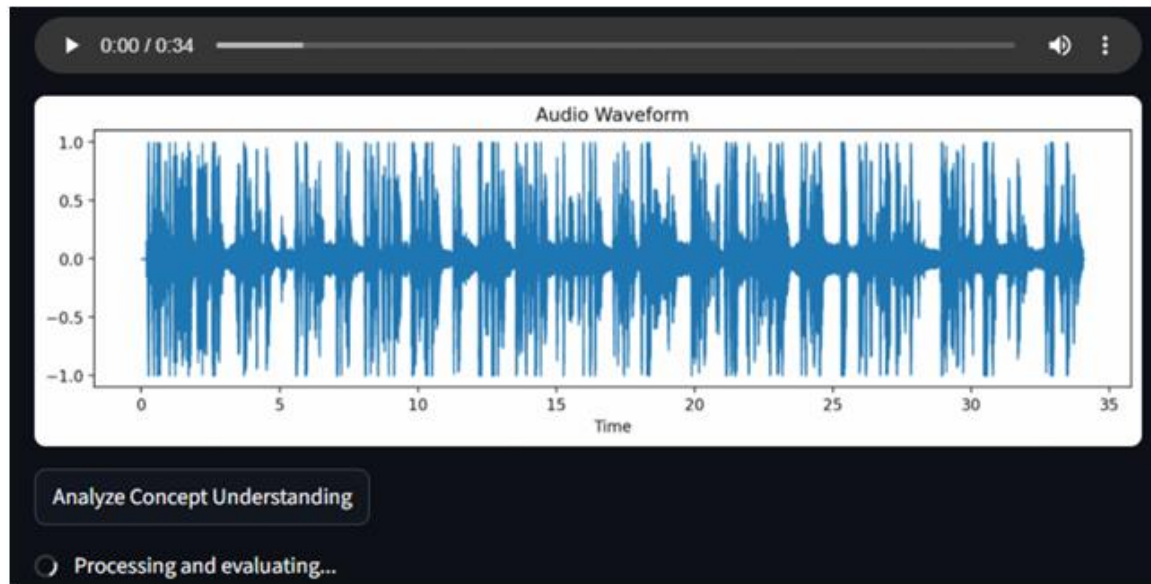
Drag and drop file here
Limit 200MB per file • WAV, MP3

Browse files

Concept Reference

Machine Learning is a subset of artificial intelligence that allows systems to learn patterns from data and improve performance without being explicitly programmed.

Upload an audio file to begin analysis.



Analysis Completed

Transcribed Explanation

If you don't discipline when you are school or in college, you pay a heavy price in later part of your life. Discipline like coming up in a proper way, writing properly, talking properly, communicating properly, respecting people properly. This really helps a long way in your life. So I don't want you should be like a potato regretting in your old age. You should remember that to be successful in life there is no shortcut. The only method is hard work and straight way and that you have to do.

Final Evaluation

Understanding Score

60/100

Moderate Understanding

Semantic Similarity	Filler Word Ratio	Confidence (Energy)
0.03	0.03	0.1633

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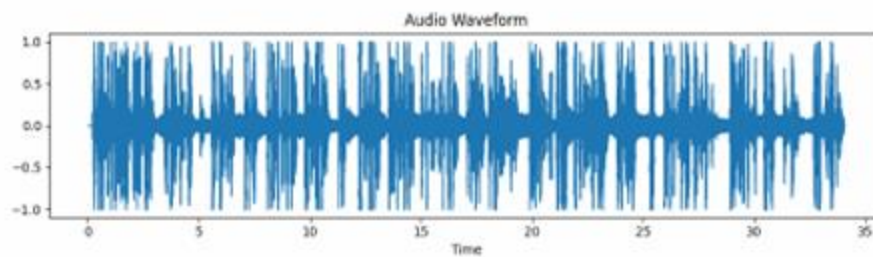
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Audio Visualization



Evaluation Summary

Metric	Value
Semantic Similarity	0.03
Filler Word Ratio	0.03
Pause Ratio	0.11
Confidence (Energy)	0.1633
Final Score	60/100
Understanding Level	Moderate Understanding